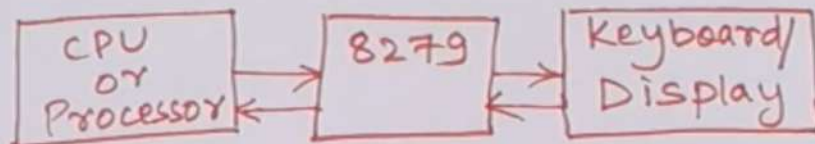


Programmable Keyboard / Display Controller (8279)



- * 8279 interfaces a keyboard with a CPU or processor.
- * 8279 scans the keyboard and identifies if any key has been pressed.
- * If it is pressed, then it sends the response to the CPU.



Features:

- * used for interfacing keyboard/display devices to microprocessor based system.
- * Simultaneous keyboard/display operations.
- * Scanned keyboard mode
- * Scanned sensor mode.
- * 8-character keyboard FIFO. 
- * 16-character display.
- * Right or left entry 16-byte display RAM.



* Programmable scan timing.



How many ways the Keyboard is Interfaced with the CPU?

* Interrupt Mode

⇒ The processor is requested service only if any key is pressed, otherwise the CPU will continue its main task.

Polled Mode.

⇒ The CPU periodically checks/reads an internal flag of 8279 to check whether any key is pressed or not.



How does 8279 work?



* Keyboard consists of maximum 64 keys which are interfaced with the CPU by using the key codes.

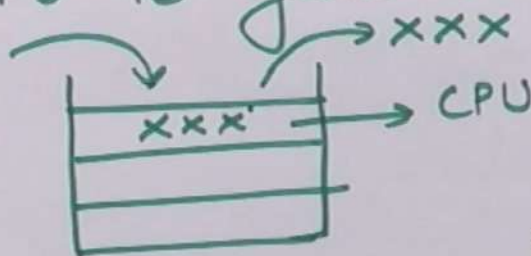
* Key codes are stored in an 8-byte FIFO RAM which can be accessed by CPU.

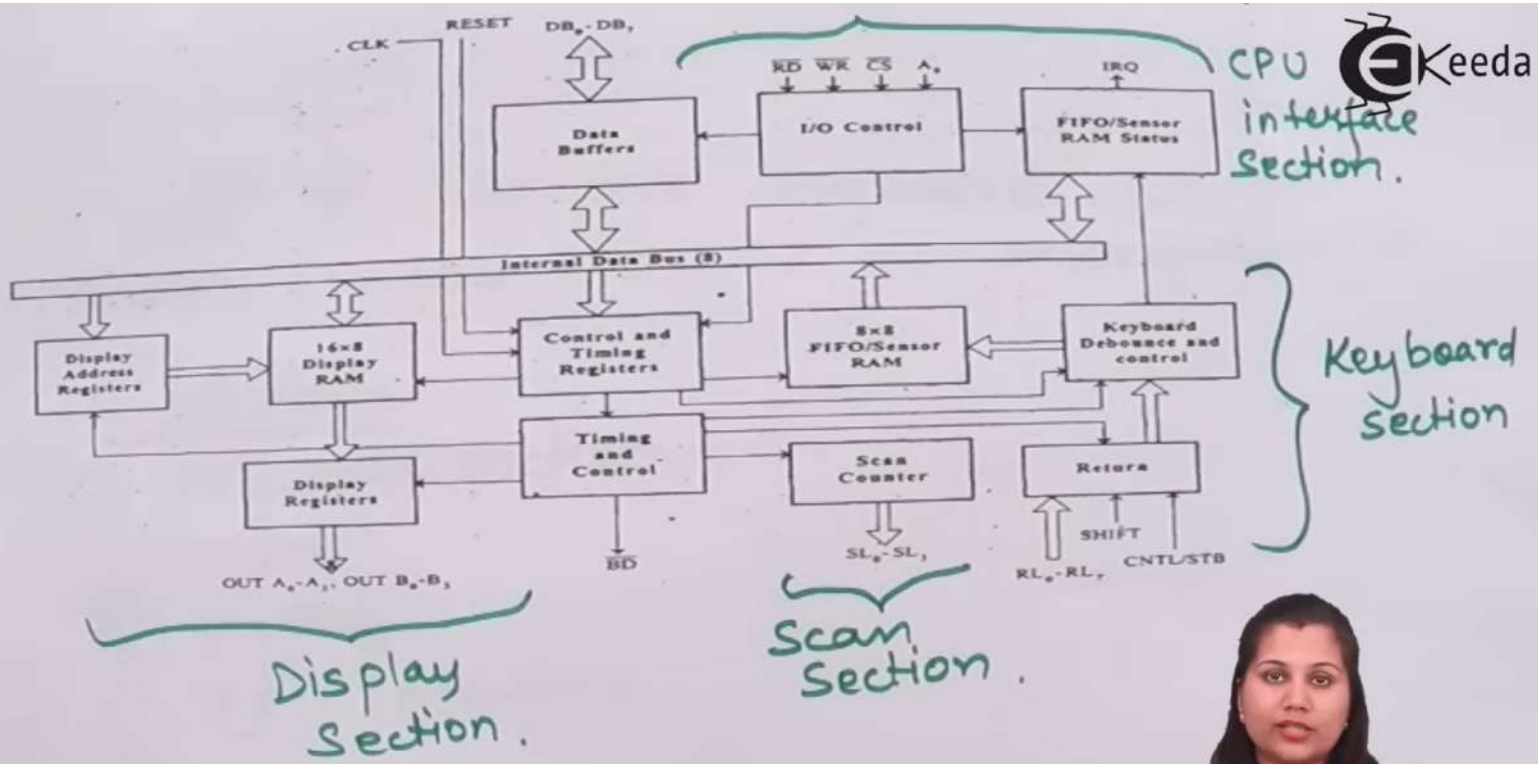
* If more than 8 keys / characters are entered in the FIFO, then it means more than 8 keys are pressed at a time. This is when OVERRUN status is set.



* If FIFO contains a valid key entry, then the CPU is interrupted in an interrupt mode else the CPU checks the status in polling to read the entry.

* Once the CPU reads a key entry, then FIFO is updated and the key entry is pushed out of the FIFO to generate space for new entries.





Programmable Keyboard / Display Controller (8279) Keeda

* used to interface keyboard and display devices to CPU or microprocessor.

4 sections

- ① Keyboard section
- ② Scan
- ③ Display
- ④ CPU interface.



Keyboard section



* Return lines RL0 - RL7. that can be used to form the columns of a keyboard matrix.

* Two additional i/p — SHIFT
 \ CONTROL / STROBE.

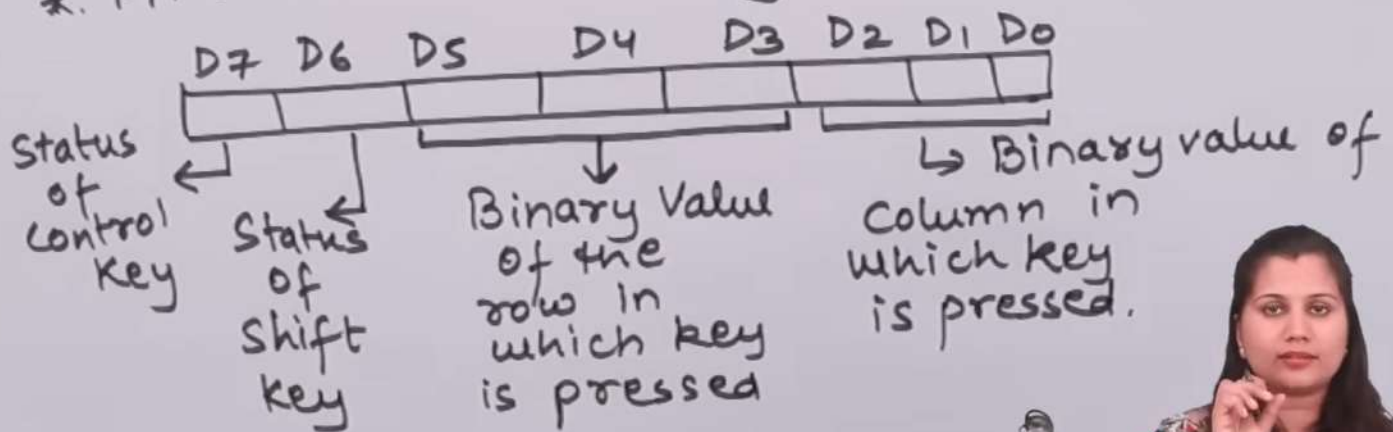
* Modes — 2 Key Lockout
 \ N-Key Roll Over.

* 8x8 FIFO Sensor RAM
First In First Out.



* FIFO RAM $\rightarrow$ where the keycode of every pressed key is entered as per their sequence.

* FIFO can store 8 key codes.



* FIFO Sensor / RAM status generate an interrupt signal when there is an entry in FIFO.

* ^{Scan} Keyboard mode — 64 Keys.

^{Scan} sensor — condition of 64 switches.

* Keyboard debounce and control unit — first scans the key closure row-wise, if found then the unit debounce the key entry.



Display Section:-

* 8 output lines divided into 2 groups
OUT A₀-A₃ OUT B₀-B₃.

* single group of 8 lines.
two groups of 4 lines.

* Display address registers holds the addresses of the word currently read / written by the CPU to / from the display RAM.



* The display can be blanked by the $\overline{BD}$ line.

* 16x8 display RAM.

* CPU can read from or write into any location of the display RAM.



Scan Section:

* Scan Counter and four scan lines
SL0-SL3.

* Decoded scan mode, the o/p of scan lines
will be similar to a 2-to-4 decoder.

* Encoded scan mode, the o/p of scan lines
will be binary count.

* Scan lines are used to form the row of
a matrix keyboard and also connected
to digit drivers of a
display.

CPU Interface section:-

- * This unit takes care of data transfer between 8279 and the processor.
- * 8 bidirectional data lines $DB_0 - DB_7$
- * It requires two internal address $A=0$ for selecting data buffer and $A=1$ for selecting control register.

* Control signals — $\overline{WR}$ (Write)
 $\overline{RD}$ (Read)
 $\overline{CS}$ (Chip Select)
read/write to 8279.



- * Interrupt Request line IRQ, for interrupt driven data transfer with processor.
- * 8279 require an internal clock frequency of 100kHz.
- * can be obtained by dividing the input clock by an internal pre-scaler.
- * RESET signal sets the 8279 in 16-character display with 2-key lockout Keyboard modes.

